

Required practical: Resistance

Year 10 GCSE Physics • AQA Trilogy RP 15 • Physics RP 3

Name _____

Class _____ Date _____

Plan. Measure. Explain.

Work in pairs. Read the whole method before you build anything.

Online version with hints: panphy.app/gcsephy/year10phy/unit01/practical/resistance/

1 Aim

Use circuit diagrams to set up circuits that investigate the factors affecting resistance: the length of a wire at constant temperature, and combinations of resistors in series and parallel.

$$R = V \div I$$

R in ohms (Ω) · V in volts (V) · I in amperes (A)

2 Apparatus

- Low-voltage d.c. power supply or battery (about 2 V)
- Ammeter (0–2 A)
- Voltmeter (0–5 V)
- Switch and connecting leads
- Two crocodile clips
- Metre ruler and sticky tape
- About 1.1 m of thin constantan or nichrome wire
- Two identical resistors, for example 10 Ω (Part B)

3 Safety

- The thin wire can get hot enough to burn. Do not touch it while the current is on.
- Switch off between readings. This also stops the wire heating up.
- Keep the p.d. low. Never connect the wire straight across the supply without the ammeter in the circuit.

Before you start

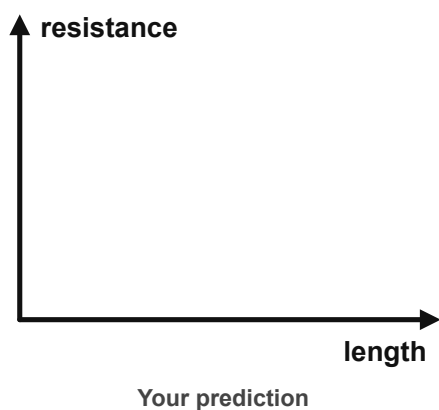
Learn the key words, then make a prediction before you build anything.

4 Key words

Independent variable	The variable you change.
Dependent variable	The variable you measure.
Control variable	A variable you keep the same, so the test is fair.
Resolution	The smallest change a meter can show, for example 0.01 A.
Anomalous result	A result that does not fit the pattern of the others.
Line of best fit	One straight line (or smooth curve) drawn as close as possible to all the points.

5 Predict

Sketch how you think the resistance of the wire will change as its length increases.



Q1 Give a reason for your prediction.

Part A Length of a wire

Part A • Resistance • Change the length of wire between two crocodile clips. Measure the p.d. and current each time and calculate the resistance.

1 Variables

Variable	What it is	How
Independent	Length of the wire between the clips	Move the sliding clip in 10 cm steps
Dependent	Resistance of the wire	Measure V and I, then calculate $R = V \div I$
Control	Material and thickness (diameter) of the wire	Use the same piece of wire throughout
Control	Temperature of the wire	Low p.d.; switch off between readings

2 Method

1. Tape the wire tightly along the metre ruler so it is straight. Line up one end with the 0 cm mark.

2. Build the circuit below. Attach the fixed crocodile clip at 0 cm and the sliding clip at 10 cm.

3. Close the switch. Read the voltmeter and the ammeter quickly, then open the switch.

4. Move the sliding clip 10 cm further along the wire and repeat step 3. Carry on up to 100 cm.

5. Calculate the resistance for each length using $R = V \div I$.

6. Repeat any reading that does not fit the pattern of the others.
- ☐

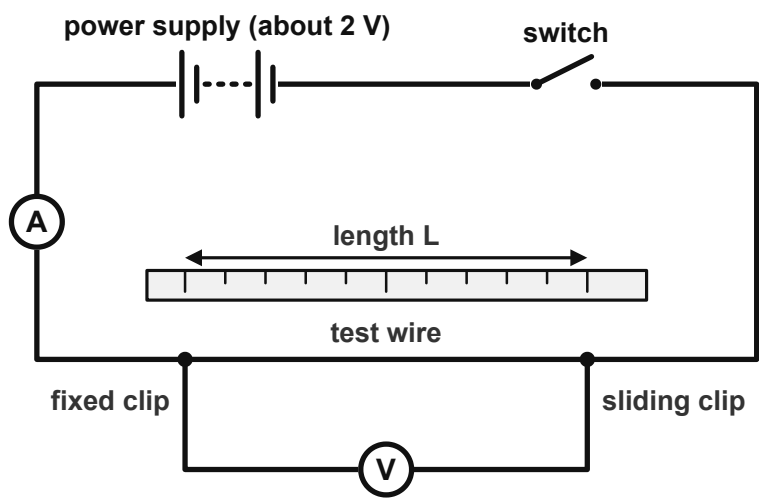
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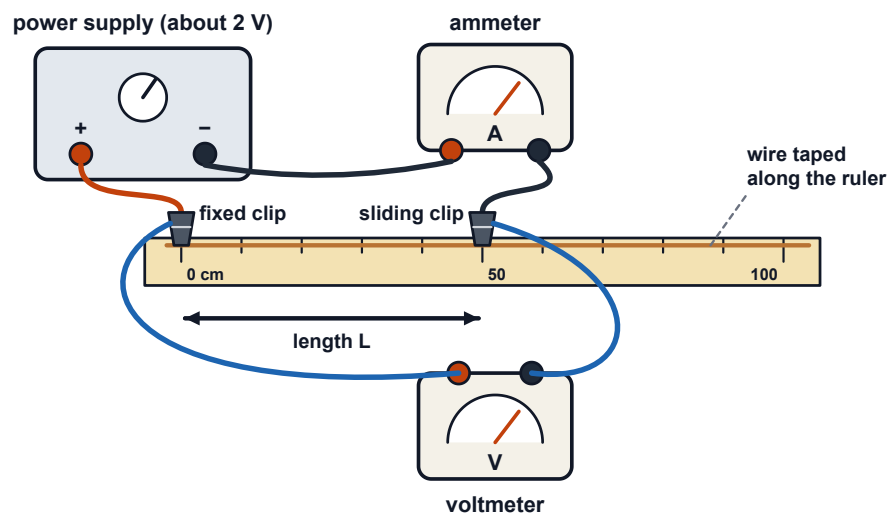
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The wire is taped along a metre ruler. The voltmeter is connected across the wire between the two crocodile clips.

Part A Length of a wire



On the bench: the ammeter is in series; the voltmeter leads clip on at the same points as the crocodile clips. Include a switch in the circuit.

Measure carefully. Measure each length to the same contact point on the crocodile clips, and read the ruler at eye level.

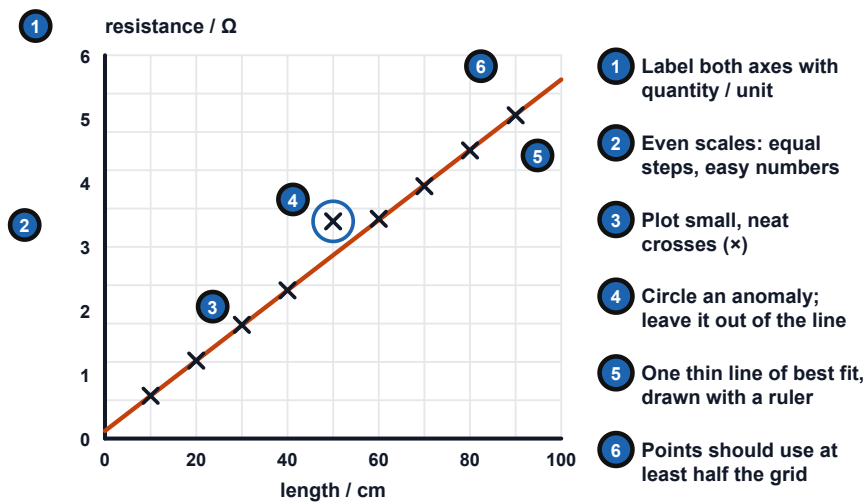
3 Results

Give p.d. and current to 2 decimal places. Give resistance to 2 decimal places.

Length / cm	p.d. / V	Current / A	Resistance / Ω
10			
20			
30			
40			
50			
60			
70			
80			
90			
100			

Part A Length of a wire

Q2 Look at your resistance column. Describe how the resistance changes as the length of the wire increases.

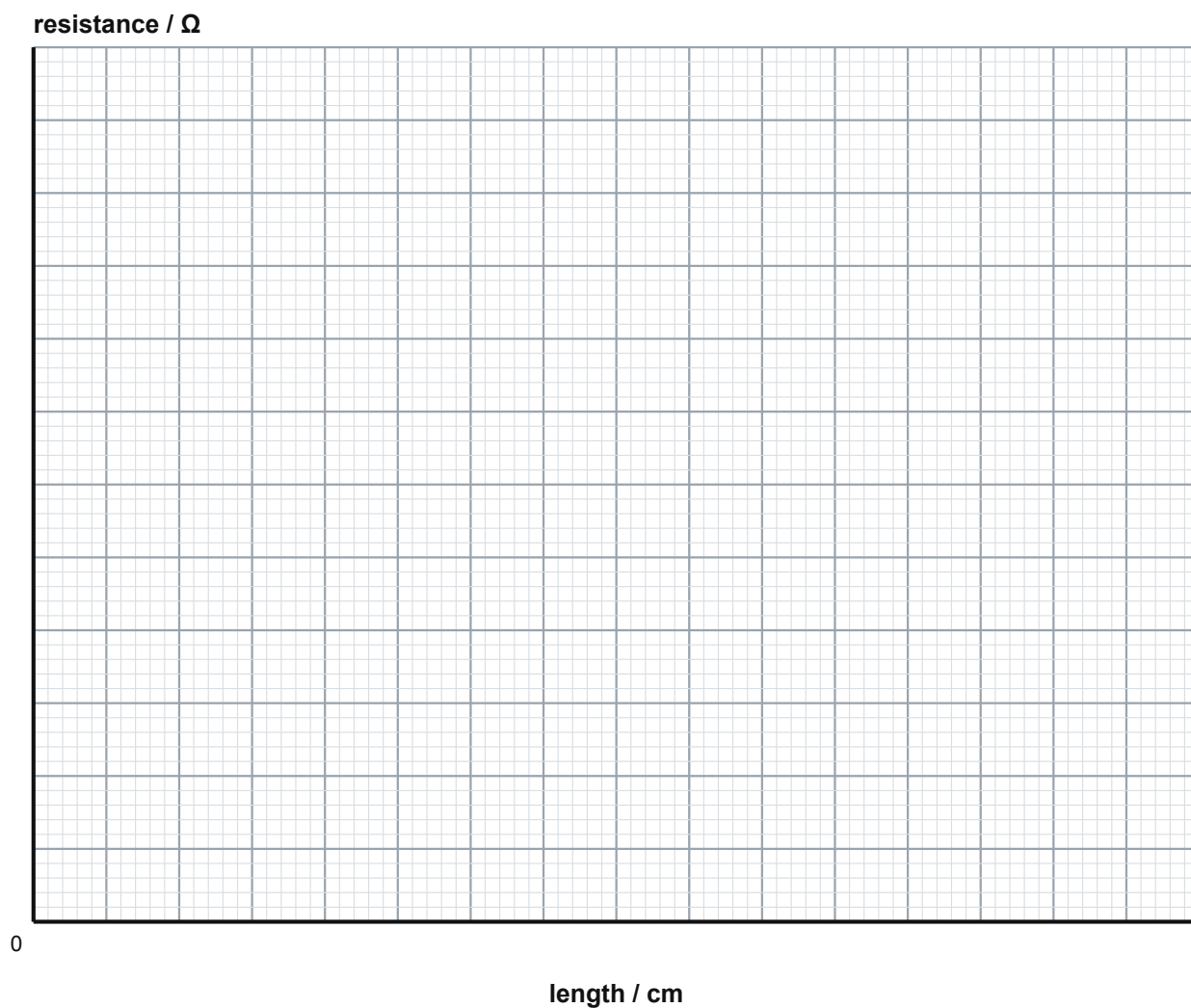


A good graph: the numbered checks are how graph marks are awarded.

Part A Length of a wire

4 Plot your graph

Plot resistance (vertical axis) against length (horizontal axis). Choose scales that use at least half of the grid. Draw a straight line of best fit.



Q3 Describe your graph. What does it show about the relationship between resistance and length?

Part A Length of a wire

Q4 Use your graph to find the resistance of 45 cm of this wire. Show on your graph how you did this.

Q5 The line of best fit may cross the resistance axis slightly above zero. Suggest why.

Q6 Explain why the circuit should be switched off between readings.

Q7 Explain, in terms of electrons, why a longer wire has a greater resistance.

Part A Length of a wire

Q8 A 30 cm length of the wire had a p.d. of 1.19 V across it and a current of 0.67 A through it. Calculate its resistance.

Equation

Substitute

Answer unit

Q9 The ammeter shows readings such as 0.67 A. What is the resolution of the ammeter?

Q10 Look at your graph. Identify any anomalous result, or state that there are none. What should you do with an anomalous result?

Q11 Suggest one improvement to the method. Explain how it would make the results better.

Part B Resistors in series and parallel

Part B • Resistance • Measure the resistance of each resistor on its own, then of the two resistors joined in series and in parallel.

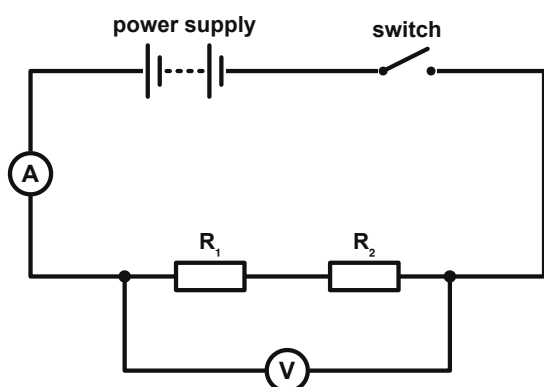
1 Variables

Variable	What it is	How
Independent	How the resistors are arranged	One resistor, two in series, two in parallel
Dependent	Total resistance of the arrangement	Measure V and I, then calculate $R = V \div I$
Control	The resistors and the supply setting	Use the same two resistors and supply throughout

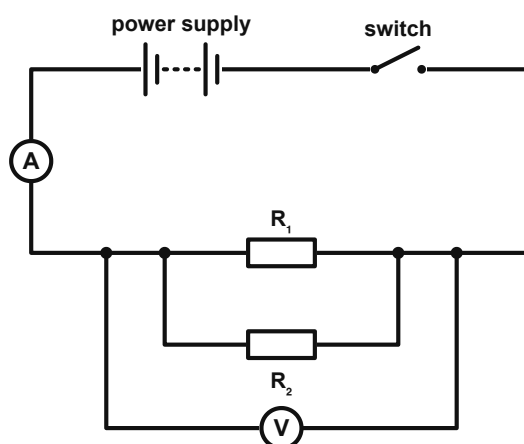
Q12 Before you measure: predict the total resistance of R_1 and R_2 in series, and in parallel. Is each one bigger or smaller than one resistor on its own?

2 Method

1. Connect resistor R_1 on its own, with the ammeter in series and the voltmeter across the resistor. ☐
2. Close the switch. Record the p.d. and the current, then open the switch. ☐
3. Replace R_1 with R_2 and repeat step 2. ☐
4. Connect R_1 and R_2 in series, one after the other. Put the voltmeter across both resistors. Repeat step 2. ☐
5. Connect R_1 and R_2 in parallel, on separate branches between the same two junctions. Put the voltmeter across the pair. Repeat step 2. ☐
6. Calculate the resistance for each arrangement using $R = V \div I$. ☐



Series: the voltmeter goes across both resistors.



Parallel: the voltmeter goes across the pair of resistors.

Part B Resistors in series and parallel

3 Results

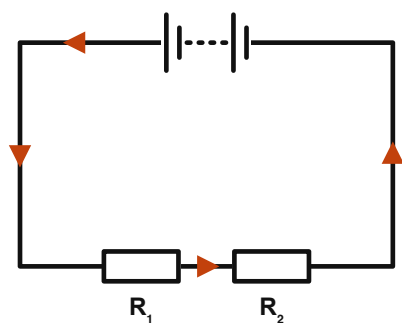
Give resistance to 1 decimal place.

Arrangement	p.d. / V	Current / A	Resistance / Ω
R_1 on its own			
R_2 on its own			
R_1 and R_2 in series			
R_1 and R_2 in parallel			

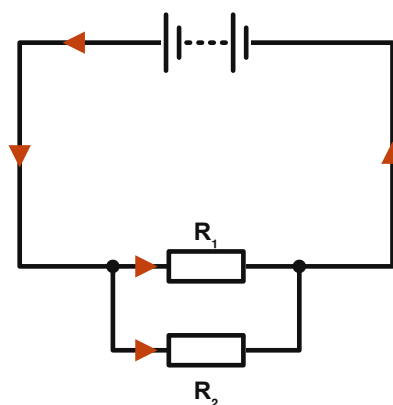
Q13 Add R_1 and R_2 from your table. Compare the answer with your measured resistance for the series arrangement.

Q14 How does the resistance of the parallel arrangement compare with the resistance of R_1 and of R_2 on their own?

Part B Resistors in series and parallel



Series: one path.
The charge goes through R_1 and then R_2 .



Parallel: two paths. More current flows
for the same p.d., so the total R is smaller.

Arrows show the conventional current. Adding a branch in parallel adds another path.

Q15 Explain why adding a resistor in parallel decreases the total resistance.

Q16 Predict the total resistance of three of these resistors in series. What would happen to the total resistance if you kept adding more of them in parallel instead?

Q17 A $22\ \Omega$ resistor and a $47\ \Omega$ resistor are connected in series to a $6.0\ \text{V}$ supply. Calculate the total resistance, then the current.

Equation

Substitute

Answer unit

Exam practice Plan the practical

Exam practice • Resistance • A six-mark method question like the ones AQA asks about this practical. Write your answer without looking at the method.

AQA-STYLE **Q18** A student wants to investigate how the length of a wire affects its resistance. Describe a method the student could use. You may include a circuit diagram in your answer. **[6 marks]**

1 Check your answer

When you have finished, tick each point your method includes.

- | | |
|--|--|
| ☐ A circuit: ammeter in series, voltmeter across the wire | ☐ Change the length in equal steps over a wide range |
| ☐ Measure the length between the clips with a ruler | ☐ Same wire; low p.d. and switch off between readings |
| ☐ Record V and I ; calculate $R = V \div I$ | ☐ Repeat readings; plot R against length |